Release and rollback

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which dependency or mandatory gate still blocks the proposal.

Teach release evidence, compatible recovery and preserved repair history without executing deployment.

Keep one habit: Model incompatible rollback → compute supplied scores → enforce privacy gate → inspect remaining gates → assess prompt/corpus dependency.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of release and **rollback**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of deployment environments

Reference example, not the only acceptable wording: Teacher explanation: Development, staging and production are distinct environments with explicit data and credential boundaries.

Test configuration must not silently target production. For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed environment names.

A staging run cannot use production write credentials.

Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of caching and stale information

Reference example, not the only acceptable wording: Teacher explanation: Caching reuses results but needs freshness and access-aware keys. A cached answer must not cross role or corpus-version boundaries.

Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.

Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of untrusted **input** boundaries

Reference example, not the only acceptable wording: Teacher explanation: Untrusted **input** must be treated according to a contract before reaching code, queries or tools.

Data that resembles instructions does not acquire execution authority. For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code.

Reject invalid length or characters.

Pictured end goal

Explain release and **rollback** using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Deployment environments

Caching and stale information

Untrusted input boundaries

Use the three-session teaching cycle above, with extra time as needed. Release records bind code, configuration, data schema and evaluation **evidence**. **Rollback** requires **compatibility** with the current stored data. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of release and **rollback**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.
</details>

MAIN LESSON · SECTION 2

Teach complete version dependencies and mandatory release gates

> Keep the main schema mismatch distinct from the picture's privacy veto.

Definitions and complete fictional manifest

Release records bind code, configuration, data schema and evaluation **evidence**. **Rollback** requires **compatibility** with the current stored data.

Version v2 writes schema 2. Version v1 reads only schema 1. A **rollback** to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.

Declared release or rule	Complete fictional classroom contract
Release A manifest	Code v1; cfg 1; schema 1; prompt p1; corpus c1; eval-A
Release B manifest	Code v2; cfg 2; schema 2; prompt p2; corpus c2; eval-B
Main reader compatibility	v1 reads only schema 1; v2 reads schemas 1 and 2
Main current inventory	Observed record schemas [1, 2, 2] after v2 writes
Main code-only proposal	Return to v1 without changing current data: block schema 2 incompatibility
Separate picture correctness	A 8/10 = 80%; B 9/10 = 90%; these cases only
Mandatory picture privacy gate	B fails; release blocked regardless of higher correctness
Code-only quality threshold	Fictional 80% threshold for the local gate model, not a universal release rule
Gate model	Explicit Boolean quality, privacy, compatibility and recovery results; all required

Declared release or rule	Complete fictional classroom contract
Limits	A compatibility check or all supplied gates passing supports review only; no real migration, adapter, deployment or external-effect undo

Main compatibility matrix and checked reasoning

Main current data schema	v1 reader supports it?	v2 reader supports it?
1	Yes	Yes
2	No	Yes
2	No	Yes
Proposed code-only v1 rollback	Blocked: schema 2 remains stored	No data conversion performed
Retain v2 reader	Passes only this declared schema check	Not a full recovery approval
Possible compatible recovery	Needs a tested compatible reader or approved migration/adapter	No such operation is executed here

Main v1 only reads schema 1, so current [1, 2, 2] blocks code-only rollback. v2 reads both schema versions but that establishes only its declared reader compatibility. Picture A is 80% and B 90%, a 10-percentage-point increase, yet mandatory privacy failure blocks B. A fresh reserved privacy pass does not discharge compatibility, recovery or every other required privacy condition.

The local four-gate model declares an 80% quality threshold only for its fiction. All supplied gates passing yields review eligibility, not automatic deployment. No recovery adapter, migration or effect reversal is executed. The selected old-prompt/new-corpus case requires restoring compatible field dependencies or a tested adapter.

Release and rollback — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
After deployment v2 creates schema 2 records; operator proposes rollback code alone to v1.	Version v2 writes schema 2. Version v1 reads only schema 1. A rollback to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.	Code-only rollback is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure evidence.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Code-only **rollback** is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

Qualification: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1.

Code-only **rollback** is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Gate and dependency probes

Probe	Expected interpretation
Old code cannot read new stored schema	Block code-only rollback
Reader supports all observed schemas	Passes that compatibility check only
Higher correctness but required privacy failure	Release blocked
Privacy repair passes one fresh case	New scoped evidence; other mandatory conditions still required
Missing gate or non-Boolean gate value	Reject incomplete decision record
All declared fixture gates pass	Eligible for review, not automatic deployment or universal assurance
Schema migration or adapter proposed	Must be separately tested and authorized; not assumed available
External action already completed	Code rollback does not automatically undo it
Old prompt needs fields absent from new corpus	Dependency mismatch; restore compatible combination or tested adapter

Separate score comparison and repair-evidence scope

A higher score can still block release. Version B is proposed to replace A; every privacy gate is mandatory.

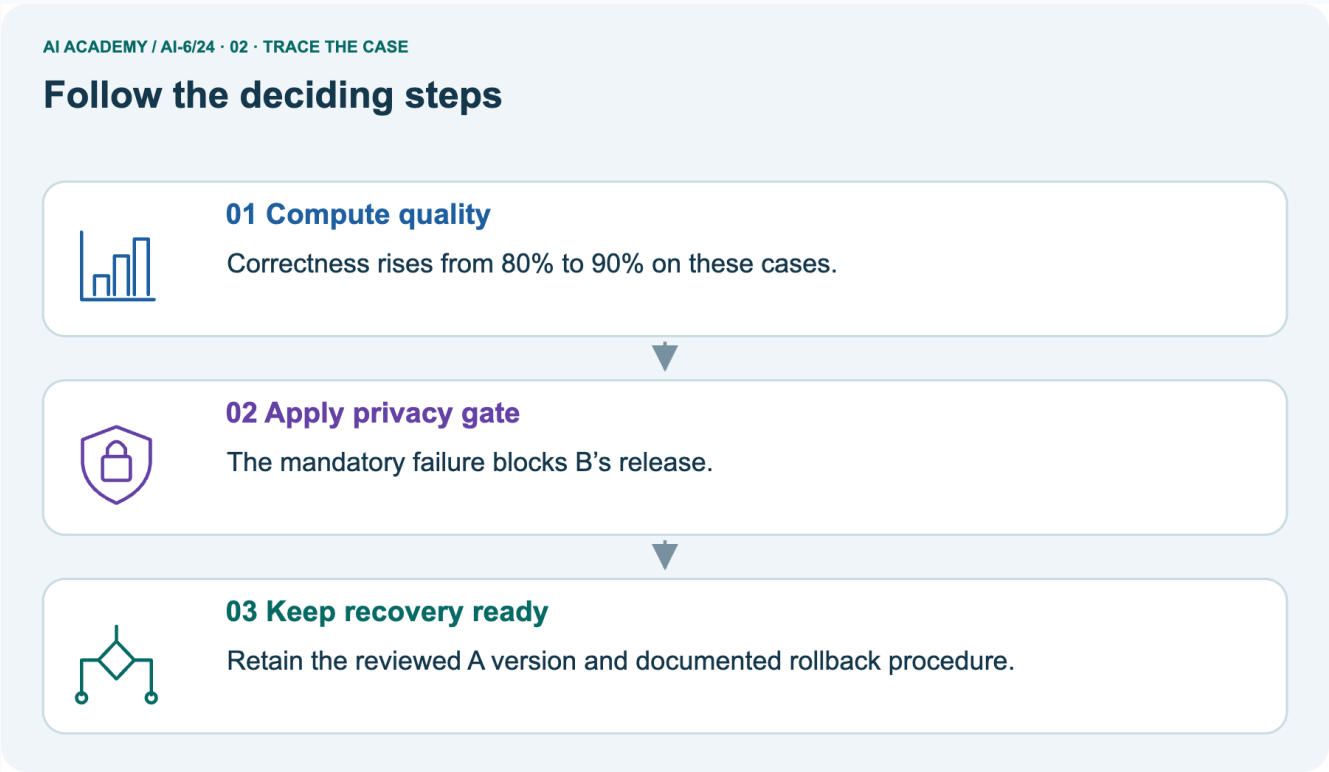
A passes 8/10 correctness cases. B passes 9/10 but fails one mandatory privacy check.

The main rollback example concerns reader compatibility with schema 2. The separate picture compares A and B on ten correctness cases: 8/10 is 80%, 9/10 is 90%, a rise of 10 percentage points. B's mandatory privacy failure still blocks release.

Repairing B and passing a fresh reserved privacy case adds evidence for that privacy check; it does not prove that every other required check passed or that all privacy risks are resolved. Keep the original failure and the repaired version's new evidence.

Retaining reviewed A and a documented recovery plan is useful only if its dependencies remain compatible with current stored data; neither rollback nor this picture reverses external effects.

Picture trace and fresh-privacy qualification



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute quality	Correctness rises from 80% to 90% on these cases.
2. Apply privacy gate	The mandatory failure blocks B’s release.
3. Keep recovery ready	Retain the reviewed A version and documented rollback procedure.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Release criteria combine benefits and mandatory conditions. **Rollback** requires known versions and compatible data, not an informal promise.

Rolling back code may not reverse earlier external effects or incompatible data changes.

Change: Repair B and pass privacy on a fresh reserved case.

Prediction: B may enter the remaining gates; one repair does not approve everything.

Reason: All required release conditions still apply.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Release manifest	Versions and artifacts included in a release
Compatibility	Whether components can correctly use one another's contracts
Rollback	Returning to an earlier compatible operating version

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Release and rollback

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Code-only **rollback** is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Release and **rollback**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Compute A's and B's correctness percentages. Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?

Correctness rises from **80%** to 90% on these cases. The mandatory failure blocks B's release. Retain the reviewed A version and documented **rollback** procedure.

Changed case: B may enter the remaining gates; one repair does not approve everything. All required release conditions still apply.

Changed-case reasoning: B may enter the remaining gates; one repair does not approve everything. All required release conditions still apply.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main v1 rollback blocks on schema 2. Picture scores are A 80% and B 90%, but B's privacy failure is mandatory. Privacy repair alone cannot prove other gates passed; all supplied fixture gates passing gives review eligibility only. Selected independent old-prompt/new-corpus case: reverting prompt alone is invalid because required old field names are absent.

Restore a compatible reviewed combination or use a separately tested adapter and record all dependencies.

Diagnose the current misconceptions

The worked result establishes every future case: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: New release changes corpus and prompt. Old prompt requires old field names absent from the new corpus. Is reverting prompt alone valid?

Teacher explanation: No. **Rollback** must restore a compatible combination or use a tested adapter. Record all version dependencies.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Release records bind code, configuration, data schema and evaluation **evidence**. **Rollback** requires **compatibility** with the current stored data. Version v2 writes schema 2. Version v1 reads only schema 1.

A **rollback** to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.

W2 Trace the worked case

After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Code-only **rollback** is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

W3 Audit the limit

A report omits this qualification: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Code-only **rollback** is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

Qualification: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

New release changes corpus and prompt. Old prompt requires old field names absent from the new corpus. Is reverting prompt alone valid?

Teacher explanation: No. **Rollback** must restore a compatible combination or use a tested adapter. Record all version dependencies.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: No. **Rollback** must restore a compatible combination or use a tested adapter. Record all version dependencies. Require intermediate reasoning and the stated scope.

Fresh retry: v3 config uses renamed key; v2 binary cannot read it. Expected reasoning: Binary **rollback** needs compatible config and validation. Version labels alone do not establish **compatibility**.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled release and **rollback evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of release and **rollback**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Monitoring quality drift. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/25, ai-6/27, ai-6/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of release and **rollback**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Code-only **rollback** is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

Check every learner before discussion: Use the worked case for Release and **rollback**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Repair B and pass privacy on a fresh reserved case. Require a **prediction** and a reason before comparing. Expected result: B may enter the remaining gates; one repair does not approve everything.

Reason: All required release conditions still apply.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Release and rollback: the case



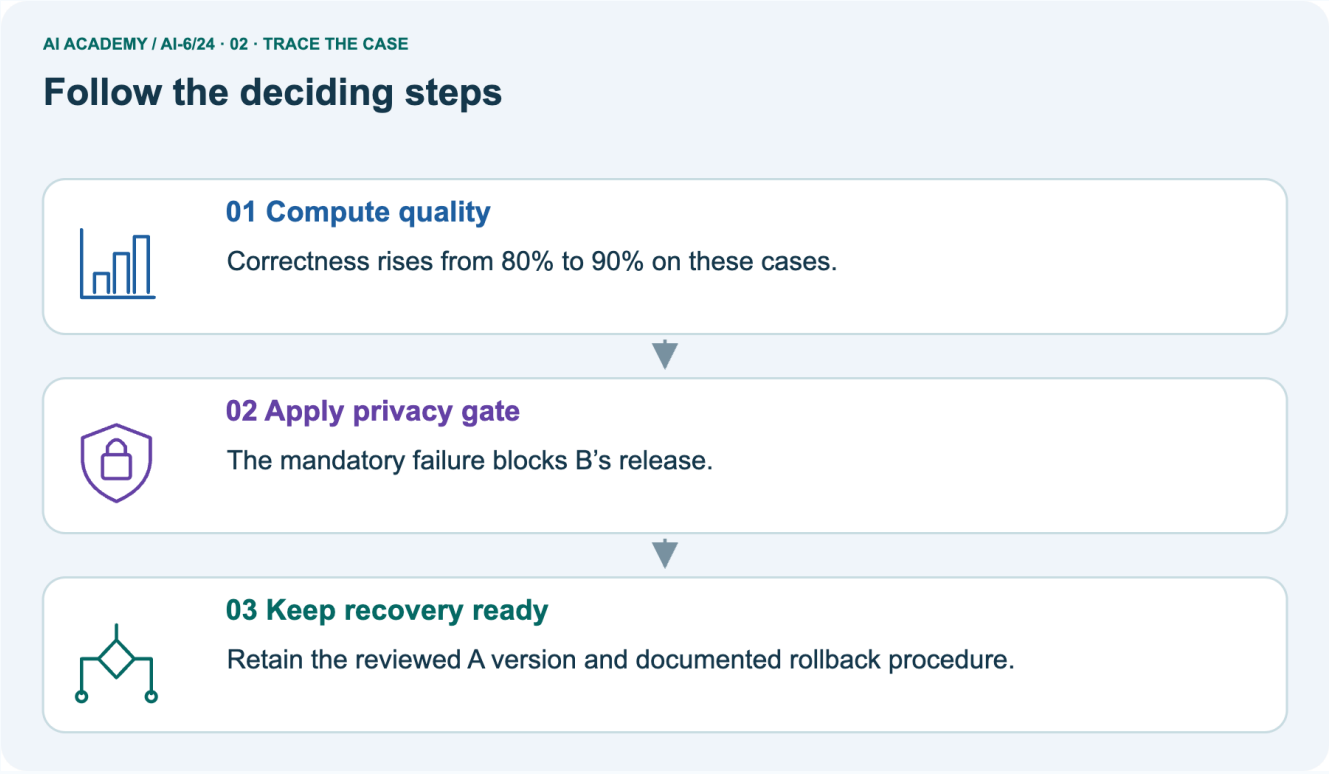
Follow the labelled picture with the matching explanation. Open the vector diagram

A higher score can still block release. Version B is proposed to replace A; every privacy gate is mandatory. Supplied facts: A passes 8/10 correctness cases. B passes 9/10 but fails one mandatory privacy check.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



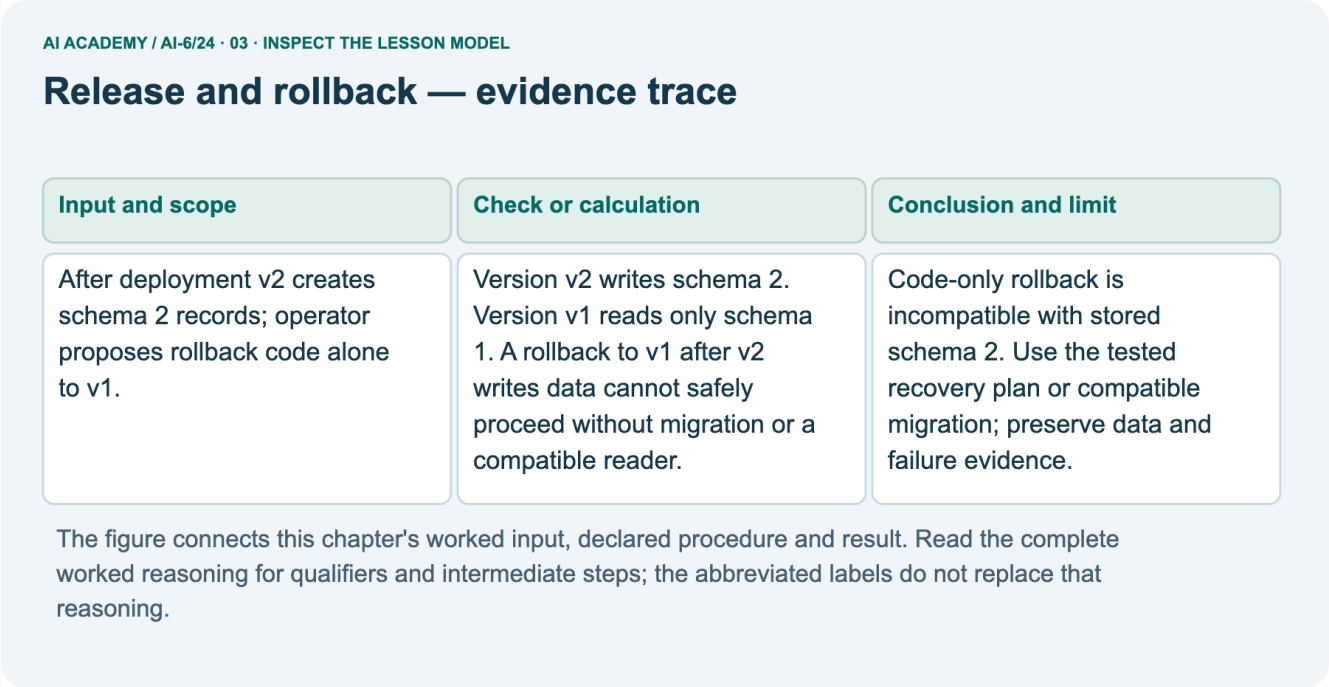
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute quality: Correctness rises from **80%** to 90% on these cases. → Apply privacy gate: The mandatory failure blocks B's release. → Keep recovery ready: Retain the reviewed A version and documented **rollback** procedure.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Release and rollback — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Release and **rollback** — **evidence** trace. After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1. | Version v2 writes schema 2. Version v1 reads only schema 1.

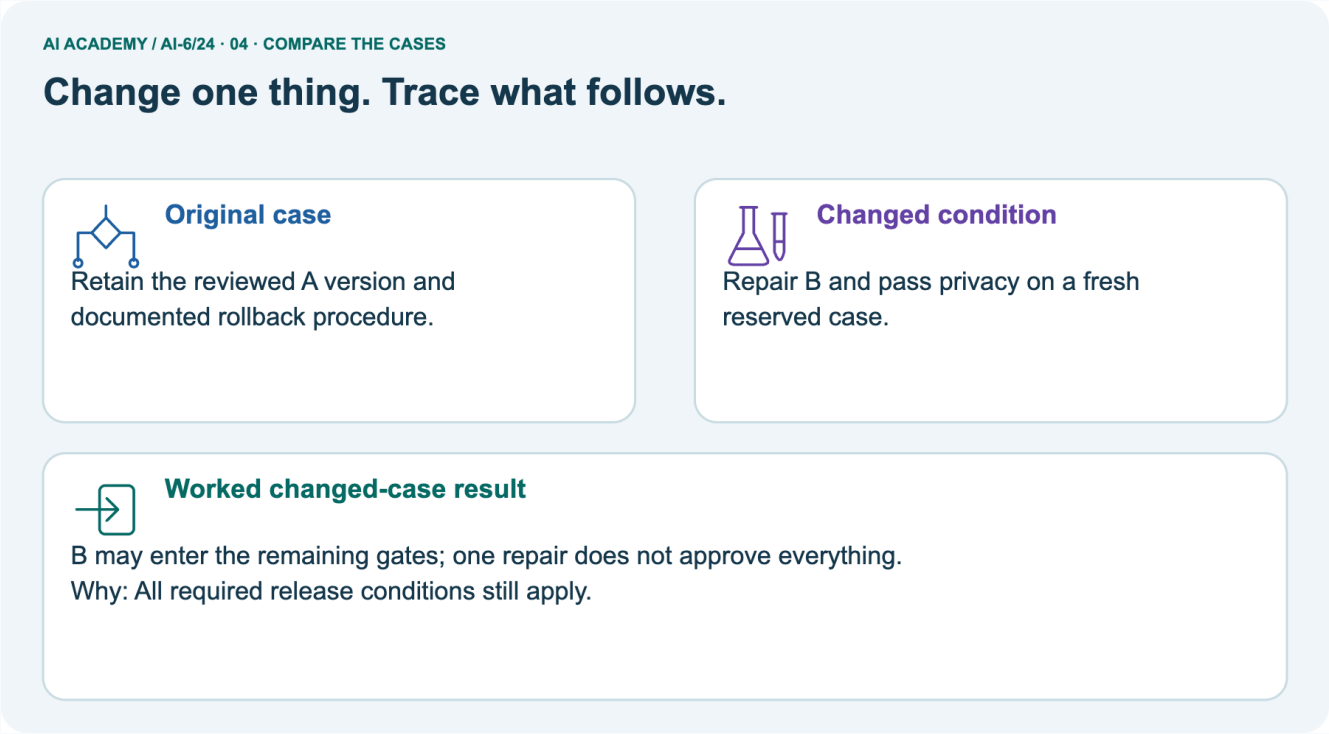
A **rollback** to v1 after v2 writes data cannot safely proceed without migration or a compatible reader. | Code-only **rollback** is incompatible with stored schema 2.

Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Retain the reviewed A version and documented **rollback** procedure. Changed condition: Repair B and pass privacy on a fresh reserved case.

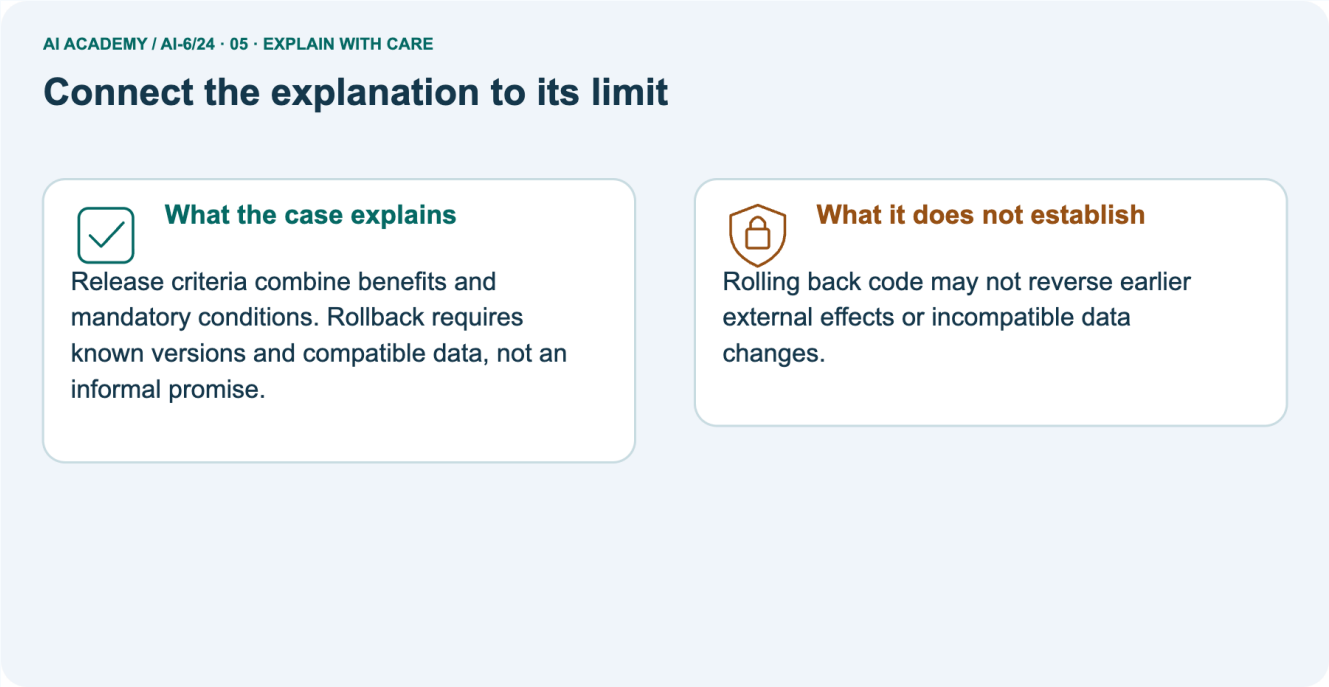
Expected result: B may enter the remaining gates; one repair does not approve everything. Reason: All required release conditions still apply.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: All required release conditions still apply.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Release criteria combine benefits and mandatory conditions. **Rollback** requires known versions and compatible data, not an informal promise.

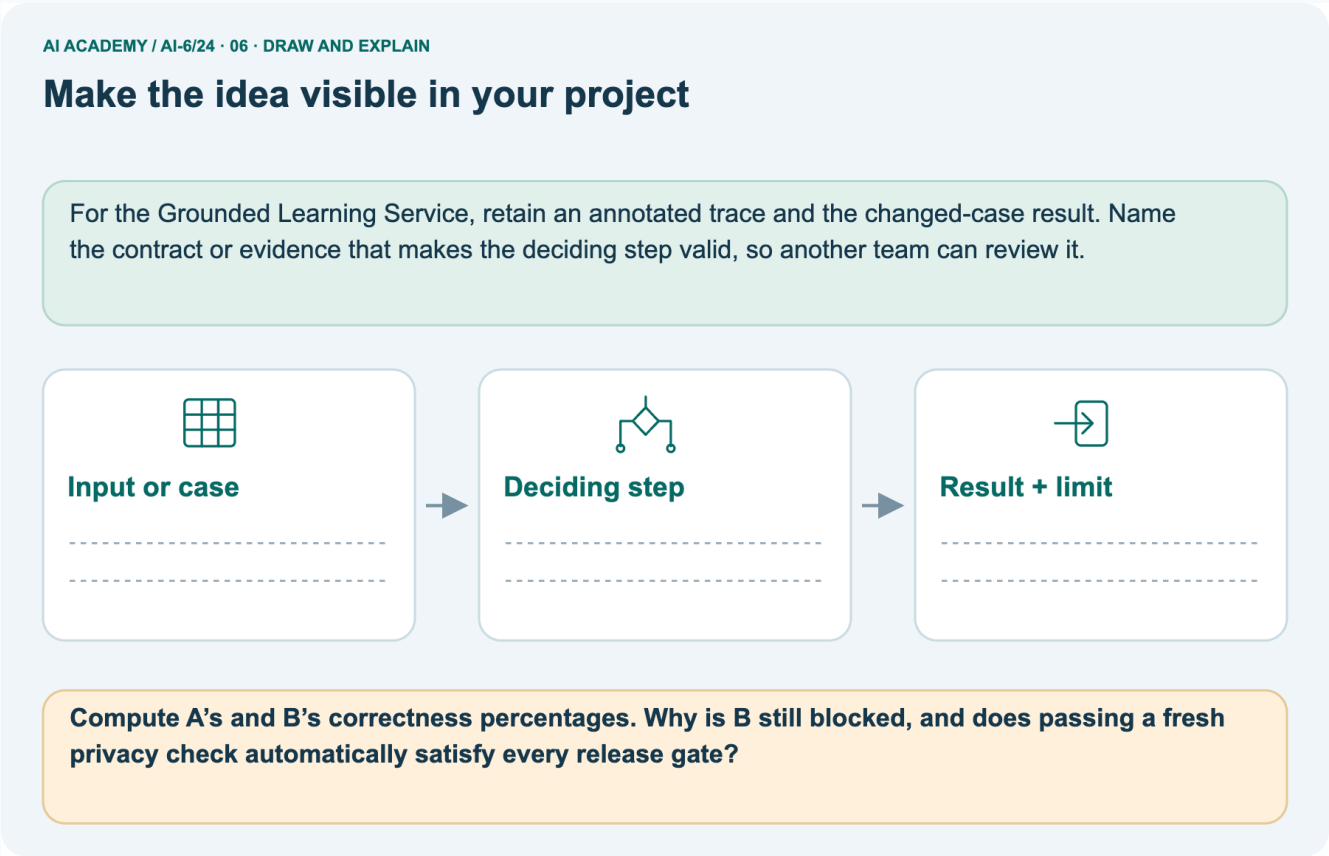
Limit: Rolling back code may not reverse earlier external effects or incompatible data changes.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Rolling back code may not reverse earlier external effects or incompatible data changes.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compute A's and B's correctness percentages. Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Correctness rises from **80%** to 90% on these cases. The mandatory failure blocks B's release. Retain the reviewed A version and documented **rollback** procedure.

Changed case: B may enter the remaining gates; one repair does not approve everything. All required release conditions still apply.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Release and rollback: the case

02. Follow the deciding steps

03. Release and rollback — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.